# University Management System

## 📌 Project Overview

This project focuses on analysing a University Management System database using SQL.

The goal is to extract meaningful insights about students, courses, professors, and academic performance to support better decision-making.

## 🎯 Objectives

- Analyse student academic behaviour and performance
- Evaluate course popularity and workload distribution
- Understand professor and department contributions
- Identify academic trends and patterns
- Perform predictive analysis for student success

## 🗂️ Database Components

The analysis is based on the following key entities:

- Students
- Courses
- Enrolments
- Professors
- Departments
- Grades

## 📊 Analysis & Insights

### 👩‍🎓 Student Analysis

**1. Students with Highest Course Enrolment**

**Insight:** Students enrolled in the highest number of courses show high academic engagement and motivation for broader learning.

**2. Consistently High-Performing Students (A/B Grades)**

**Insight:** These students demonstrate strong academic consistency and are ideal candidates for merit-based scholarships.

**3. Average Grade per Student**

**Insight:**

Helps identify:

- Top performers
- Students needing academic support & Useful for personalized counselling and mentoring.

## 📚 Course & Enrolment Analysis

**1. Top 5 Most Enrolled Courses**

**Insight:**

Popular courses likely:

- Have high career relevance
- Are core subjects
- Or are perceived as easier to score

**2. Low Enrolment Courses**

**Insight:**

- Low demand may indicate:
- Lack of awareness
- High difficulty level
- Low practical relevance

**3. Average Credits per Semester**

**Insight:**

- Balanced credits → Healthy workload
- High credits → Risk of student stress/burnout

## 👩‍🏫 Professor & Department Insights

**1. Professors Teaching Most Courses**

**Insight:** Indicates Higher workload or high student preference/popularity.

**2. Departments with Most Professors**

**Insight:** Shows resource allocation and should align with student enrolment size.

**3. Professors with Highest Average Grades**

**Insight**: Could indicate Effective teaching methods OR lenient grading

## 📈 Academic Performance & Trends

**1. Average Grade by Department**

**Insight:**

Departments with higher averages may reflect:

- Better teaching quality
- Or easier evaluation patterns

**2. Most Difficult Courses**

**Insight:**

- Courses with low average grades may need Curriculum improvement Better teaching support.

**3. Grade Trends Over Time**

**Insight:**

- Improvement → Better learning/adaptation
- Decline → Possible gaps in teaching or curriculum

## 🔍 Comparative & Predictive Analysis

**1. Department Enrolment Growth**

**Insight:**

- Fast-growing departments indicate:
- High market demand
- Student interest trends

**2. At-Risk Students Prediction**

**Insight:**

Students with:

- Low grades
- High course load

**3. Credits vs Grades Relationship**

**Insight:**

- Positive correlation → Strong students manage workload well
- Negative correlation → Overloading impacts performance

## 🛠️ Tools & Technologies

- SQL (MySQL)
- Database Queries (JOIN, GROUP BY, Aggregations, Window Functions)

## 🚀 How to Use

- Import the database schema (University Management System_DB Creation + University Management System Data Analysis)
- Run SQL queries provided in the project (University Management System Data Analysis).
- Analyse outputs and compare with insights
- Extend queries for deeper analysis

## 📌 Conclusion

This project demonstrates how SQL can be used to:

- Transform raw academic data into actionable insights
- Support decision-making in educational institutions
- Improve student performance and resource management.

## 📬 Future Improvements

- Add data visualizations (Python / Power BI / Tableau)
- Build dashboard for real-time insights
- Apply machine learning for advanced predictions